HACKATHON

# Bio-Agentic Lab Assistant

Enhancing Scientific Reasoning, Not Replacing Judgment.

**Clinical & Biological Research Orchestration.**

# The Challenge

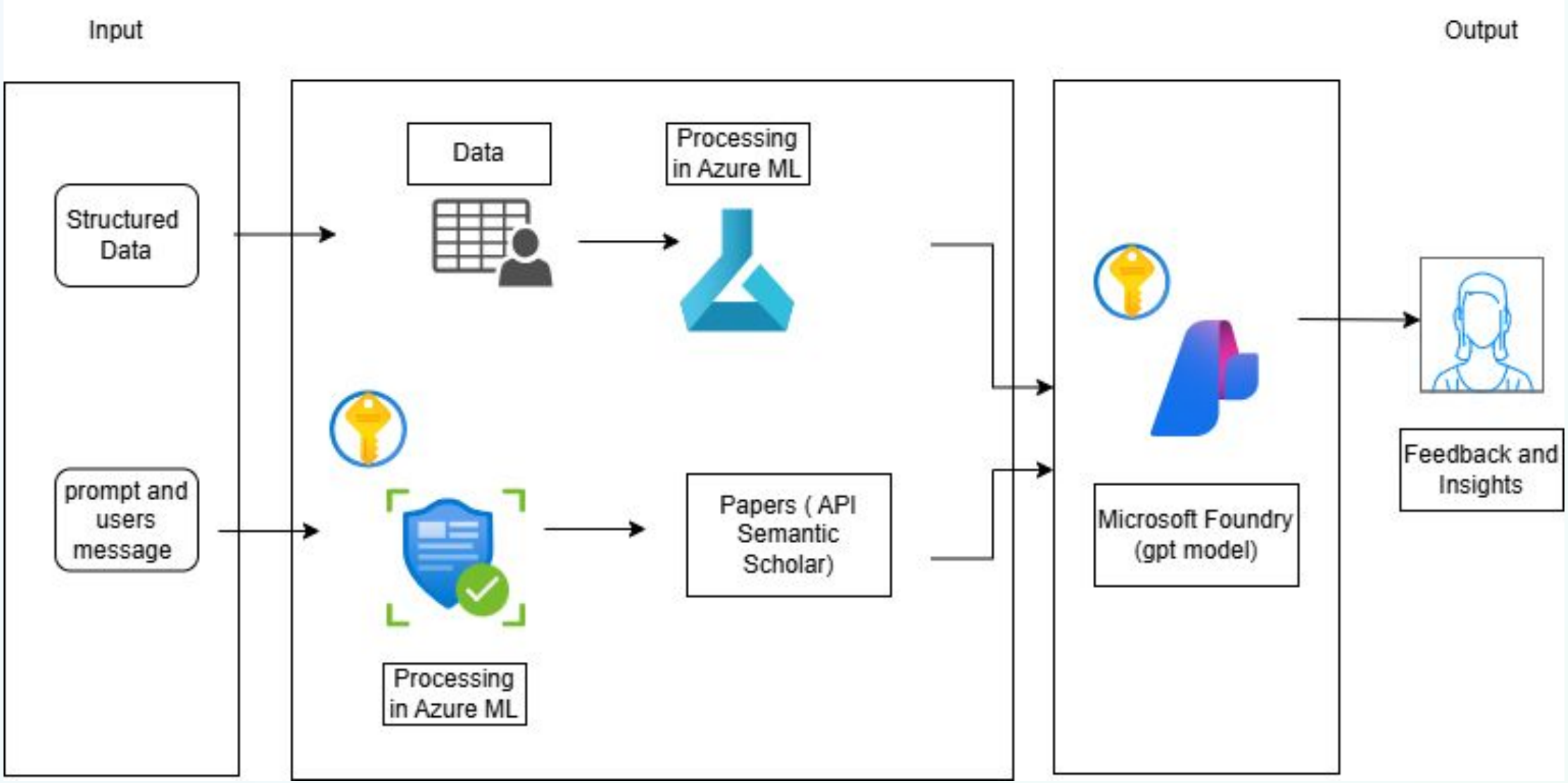
Complexity in modern biological and clinical research.

# Unleashing Scientific Curiosity

## Empowering the Lab

Researchers today are overwhelmed by manual data processing and isolated variables. Our agentic system bridge the gap:

- **Interprets:** Bio-protocols from text.
- **Suggests:** Variations for clinical next steps.
- **Analyzes:** Complex CSV results with expert "rationale".


Input
Output
Structured Data
prompt and users message
Data
Processing in Azure ML
Papers ( API Semantic Scholar)
Processing in Azure ML
Microsoft Foundry (gpt model)
Feedback and Insights

# Azure Service Breadth (25% Score)

## Data & Ingestion

**Integration between Frontend and Azure ML** for raw/clinical data zones and orchestration.

## MLOps & Intel

**Azure ML SDK v2** for pipelines, **MLflow** for tracking, and **GPT-4o** for reasoning.

## Governance

**Azure Key Vault** for all secrets.

# Responsible AI

**Safety-First Architecture:** Strict biological and clinical safety limits are not a feature; they are the core.

**Pathogen Filtering:** Pre-LLM layer blocks content related to pathogens and synthesis control.

**Non-Advisory Guardrails:** Strictly avoids forbidden medical advice behaviors.

# Performance: The MLOps Lifecycle

**0.05**

Fairlearn Bias Gate (Threshold)

## Stability in Production

We don't just deploy; we monitor clinical data drift in real-time to ensure scientific accuracy.

- **Dynamic Training:** Models retrain on user data specific to the lab.
- **Drift Detection:** Monitoring Population Stability Index (PSI) in Production.
- **Human Gate:** Equalized Odds checks before any deployment.

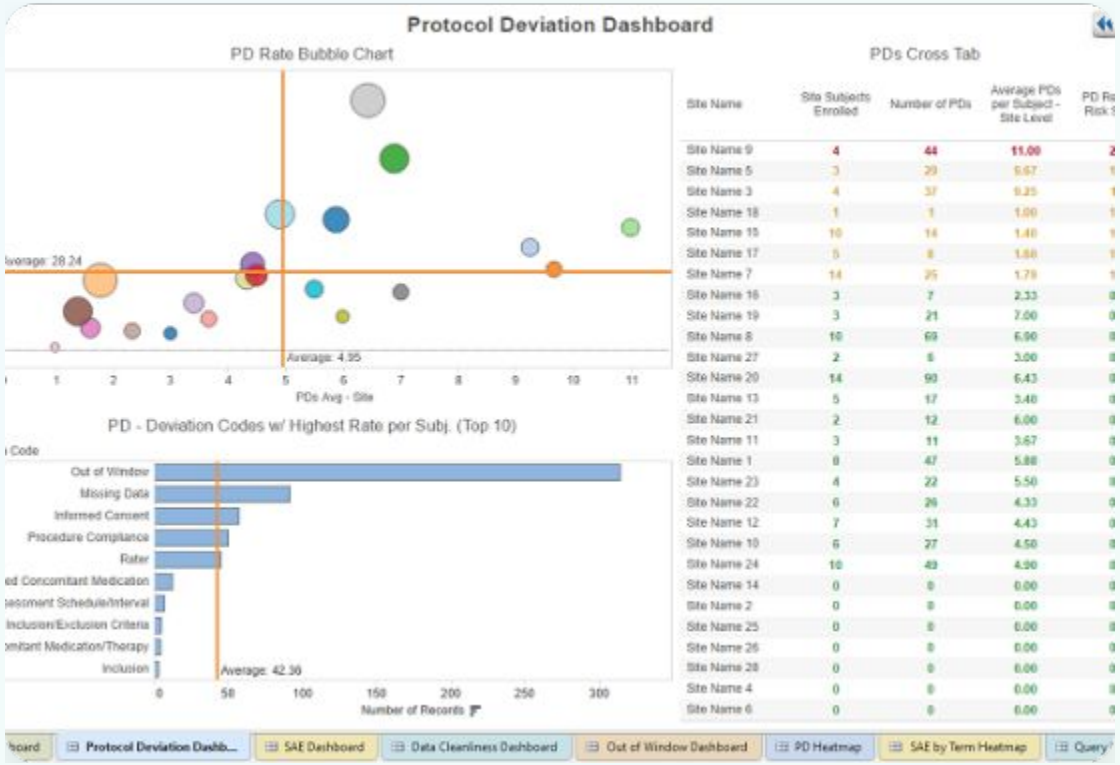
# Dual-Perspective Interface

## The "Scientist vs Tech" Toggle

A unified UI that adapts to the user's role without losing context or security.

**Scientist View:** Plain language, easy visuals, "Next Step" suggestions, and Literature sidebar.

**Technical View:** MLflow runs, AUC-ROC metrics, drift reports, and dataset hashes.



# Innovation: The Multi-Variable Advantage

## Multi-variable Reasoning

Unlike competitors that model one variable, we reason over entire complex datasets simultaneously.

## Literature Enrichment

Not just raw data—we provide "Related Research Context" connecting results to the global scientific community.